

Improvised FOC and Speed Estimation of PMSM using Reinforcement Learning and Neural Networks.

Link: <https://ieeexplore.ieee.org/document/10047821>

Abstract:

A sensor less field-oriented control of an Interior PMSM using Artificial Neural Networks (ANN) combined with Proportional–Integral (PI) control and Reinforcement Learning (RL). The performance of both methods is compared to achieve a high-speed and efficient motor response.

A deep neural network is trained to estimate rotor position and speed, eliminating physical sensors and reducing system complexity and cost. Reinforcement learning is implemented using the Twin Delayed Deep Deterministic Policy Gradient (TD3) algorithm, which employs actor–critic networks to optimise control policies by maximising long-term rewards.

Key System Components:

- **Reinforcement Learning:** A Twin Delayed Deep Deterministic Policy Gradients (TD3).
- **ANN:** 4 hidden layers with tanh in the initial and ReLU in the latter.
- **PI Controller:** with specific gain and integral value for this application.

Challenges:

The first: What RL method best suits our application? A reinforcement learning with high accuracy and a reduced variance during training will be the right choice. **Action:** One such method that satisfies the needs was the TD3, as it solves the overestimation problem present in DDPG. Adding clipped noise to actions improves robustness and reduces variance during training.

The Second: An ANN architecture needs to be built with computational weight and accuracy in mind. **Action:** The Neural network architecture consists of 4 neurons as inputs, and following that, 4 hidden layers are constructed. Adam optimiser is used to train the neural network.

The third: What parameters can yield a better response? **Action:** RL Model in a PMSM was simulated with various parameters by modifying the Layers, Neurons, Rewards, and Optimizers. A model with an increased number of Neurons has shown a better response result when compared to the other methods carried out in the same environment, which reached the reference speed 10-25ms earlier than other models.